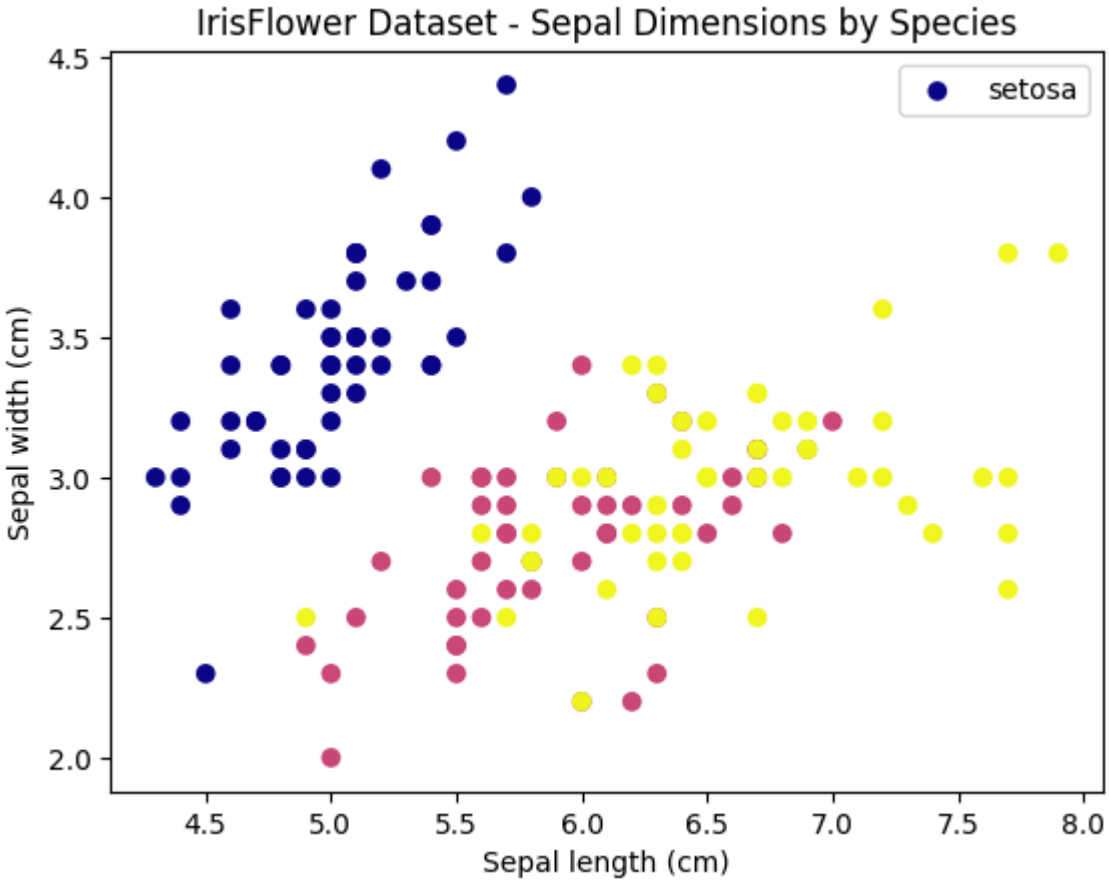
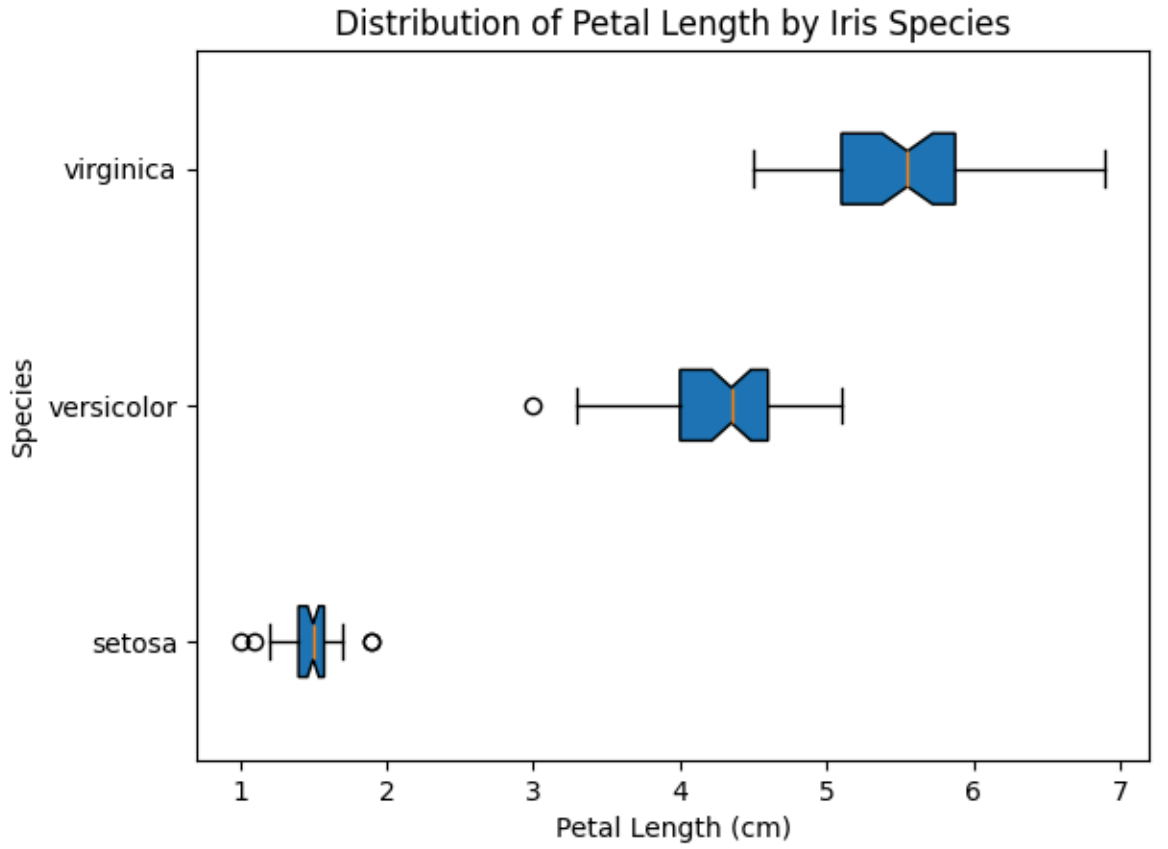
In [4]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris # To load the Iris dataset
# Data Extraction (using sklearn.datasets)
iris = load_iris() # Load the Iris flower dataset
# Data Transformation (Optional)
# The Iris data is pre-processed and ready for use
# Data Visualization (Multiple Examples)
# Example 1: Sepal Length vs. Sepal Width colored by Species (Scatter Plot)
sepal_length = iris.data[:, 0]
sepal_width = iris.data[:, 1]
target_names = iris.target_names
# Get species names
plt.scatter(sepal_length, sepal_width, c=iris.target, cmap='plasma')
# Color by target species
plt.xlabel('Sepal length (cm)')
plt.ylabel('Sepal width (cm)')
plt.title('IrisFlower Dataset - Sepal Dimensions by Species')
plt.legend(target_names)
plt.show()
```



In [5]:
```
# Example 2: Distribution of Petal Lengths across Species (Box Plot)
petal_length = iris.data[:, 2] # Petal length feature
plt.boxplot([petal_length[iris.target == 0], petal_length[iris.target == 1], pet
labels=target_names) # Separate boxes by species
plt.xlabel('Petal Length (cm)')
plt.ylabel('Species')
plt.title('Distribution of Petal Length by Iris Species')
plt.show()
```

```
C:\Users\Prasad\AppData\Local\Temp\ipykernel_10024\2513658690.py:3: MatplotlibDep
recationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  plt.boxplot([petal_length[iris.target == 0], petal_length[iris.target == 1], pe
tal_length[iris.target == 2]], notch=True, vert=False, patch_artist=True,
```



In [ ]: